



UNIVERSITY OF SRI JAYEWARDENEPURA

Faculty of Technology

Department of Information and Communication Technology

ITC 2252 – Introduction to Machine Learning

Assignment 03

Assignment 03 continues the same scenario given in Assignment 02. Therefore, before writing the answer, please go through the Answer sheet of Assignment 02.

1. Read the file " Pre_processed_data.pkl" and create the dataframe.
(If you are not satisfied with the data file you created for Assignment 02, please use the given dataset provided with this assignment.)
2. When check the count of two classes Good and ColorUnevenness , there is imbalance data. Further, there is a big difference in the number of data in Good and ColorUnevenness. We will deal with down sampling below. Reduce the number of Good by random sampling so that the ratio with ColorUnevenness is 7: 3.
(This part is not use to evaluation because code is already given in lecture note.)
3. Please convert Good to 0, ColorUnevenness to 1.
Hint: Use “pd.get_dummies” command, where “pd” is pandas
4. Delete the unnecessary data/features, drop the 'No', 'RESULT_QUALITY', 'INDT', 'R_DIECD',
5. Please divide the data into training data and test data at a ratio of 7: 3.
Hint: use “train_test_split” function
6. Apply the random forest classification model to train the dataset. Does it need the standardization technique for variables? Please use it if necessary. Otherwise, state the reason why do you think that it is not required.
7. After executing the above, verify the accuracy of prediction for test data. Then display the

accuracy for each of **training data and test data** based on following performance evolution metrics

- A. Accuracy
 - B. ConfusionMatrix
 - C. Precision
 - D. 4.Recall
 - E. F1 score
8. Using the feature selection technique, Find the top10 meaningful features, and visualize in the graph

Deadline: Please submit on or before 25.11.2025